MARTIAL WORLD

A Mini Project Report

submitted by

SARAN S KRISHNAN (MES24MCA2049)

to the APJ Abdul Kalam Technological University
in partial fulfilment of the requirements for the award of the Degree

of

Master of Computer Applications

Department of Computer Applications

MES College of Engineering

Kuttippuram, Malappuram – 679582

November, 2025

Declaration

I undersigned hereby declare that the project report MARTIAL WORLD submitted for partial fulfilment of the requirements for the award of degree of Master of Computer Applications of the APJ Abdul Kalam Technological University, Kerala, is a bonafide work done by me under supervision of <Balachandran, Associate Professor, Department of Computer Applications. This submission represents my ideas in my own words and where ideas or words of others have been included, I have adequately and accurately cited and referenced the original sources. I also declare that I have adhered to ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in my submission. I understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not been previously formed the basis for the award of any degree, diploma or similar title of any other University.

<Put Your Signature>

Saran S Krishnan(MES24MCA2049)

06/10/2025

DEPARTMENT OF COMPUTER APPLICATIONS

MES COLLEGE OF ENGINEERING, KUTTIPPURAM

CERTIFICATE

This is to certify that the report entitled **MARTIAL WORLD** is a bonafide record of the Mini Project work during the year 2025-26 carried out by S**ARAN S KRISHNAN(MES24MCA2049)** submitted to the APJ Abdul Kalam Technological University, in partial fulfilment of the requirements for the award of the Master of Computer Applications, under my guidance and supervision. This report in any form has not been submitted to any other University or Institution for any purpose.

Internal Supervisor

Head of The Department

Acknowledgment

I would like to express my sincere gratitude to my internal supervisor, Balachandhran Sir, for their invaluable guidance, support, and encouragement throughout the development of this mini project. I also extend my thanks to the Head of the Department and all faculty members of the Department of Computer Applications, MES College of Engineering, for providing the necessary resources and a conducive environment for learning and innovation.

I am deeply grateful to my fellow martial artists and peers who inspired the vision behind *Martial World*. Their insights and experiences shaped the direction of this project. Finally, I thank my family and friends for their unwavering support and motivation.

Saran S Krishnan (MES24MCA2049)

Abstract

Martial World is a pioneering web application designed to unify the global martial arts community through a dedicated digital ecosystem. It addresses the fragmentation in existing platforms by integrating commerce, social networking, and personal progress tracking tailored specifically for martial artists. The app offers a curated marketplace for gear and nutrition, a discipline-specific media-sharing network, and tools for tracking training, belt promotions, and competition history.

Built using React Native for cross-platform development and Django for backend services, Martial World leverages Firebase and AWS for real-time updates and secure hosting. This report outlines the motivation, objectives, system architecture, and implementation methodology of the project, culminating in a robust solution that empowers martial artists to grow, connect, and thrive.

Table of Contents

List of Figures ..... **vii**
List of Tables ..... **viii**
Chapter 1. Introduction **1**
1.1 Motivation 1
1.2 Objectives 2
1.3 Contributions 2
1.4 Report Organization 2
Chapter 2. System Study **3**

2.1 Existing System 3
2.2 Proposed System 3
2.3 Functionalities of Proposed System 4
Chapter 3. Methodology 5
3.1 Introduction 5
3.2 Software Tools 5
3.2.1 Python ..... **Error! Bookmark not defined.**
3.2.2 Django ..... **Error! Bookmark not defined.**
3.3 Module Description 5
3.3.1 Name of the first Module ..... **Error! Bookmark not defined.**
3.3.2 Name of the second Module ..... **Error! Bookmark not defined.**
3.4 User Story 15
3.5 Product Backlog 15
3.6 Project Plan 15
3.7 Sprint Backlog 15
3.8 Database Design 15
Chapter 4. Results and Discussions 17
4.1 Results 17
Chapter 5. Conclusion 24
References 25
Appendix 26
Appendix A Data Flow Diagram 26
Appendix B ER Diagram 27
Appendix C Source Code 28
Appemdix D Screenshot 16

List of Figures

Figure 1.1: Dialog box for creating a caption for figure using Insert → Caption option [1] ..... **Error! Bookmark not defined.**

Figure 1.2: Dialog box for creating a new source in the Reference section**Error! Bookmark not defined.**

Figure 4.1: Landing Page [4] 17

Figure 4.2: Patient profile page 18

Figure 4.3: Patient payment receipt 18

List of Tables

Table 1.1: Table caption should be placed above the table ..... **Error! Bookmark not defined.**

Table 3.1: List the software tools or languages used for the project development..... 5

Chapter 1. Introduction

Martial arts is not just a sport—it is a philosophy, a discipline, and a lifestyle embraced by millions worldwide. Despite its rich cultural heritage and widespread practice, the digital infrastructure supporting martial artists remains fragmented. Existing platforms often dilute martial arts content with unrelated material, making it difficult for practitioners to find authentic gear, track progress, or engage with like-minded individuals.

Martial World aims to bridge this gap by offering a unified platform exclusively for martial artists. It combines e-commerce, social networking, and personal development tools into a single application. The platform is designed to support practitioners across disciplines like Karate, Taekwondo, Muay Thai, Jiu-Jitsu, and Kung Fu.

The project falls under the domain of full-stack web and mobile development, with a focus on user-centric design and scalable architecture. The UI/UX is crafted to be intuitive, visually engaging, and responsive across devices, ensuring accessibility for users of all skill levels.

The problem addressed is the lack of a dedicated digital space for martial artists. Existing solutions are either too generic or fail to integrate commerce, community, and progress tracking effectively. *Martial World* offers a holistic solution that celebrates martial arts culture and supports practitioners' growth.

1.1 Motivation

As a martial artist, I have personally experienced the challenges of navigating multiple platforms to meet my training, gear, and community needs. The absence of a unified digital space that understands and caters to the unique lifestyle of martial artists inspired the creation of *Martial World*. This project is not just a technical endeavor—it is a personal mission to empower martial artists through technology.

The motivation stems from the desire to build a platform that reflects the values of discipline, growth, and community inherent in martial arts. By integrating commerce, content, and progress tracking, *Martial World* aims to simplify and enrich the journey of every practitioner.

1.2 Objectives

The primary objective of *Martial World* is to develop a mobile application that:

- Empowers users to track and analyze their martial arts journey.
- Fosters a vibrant, discipline-specific social network.
- Centralizes martial arts-related commerce through a curated marketplace.

This project builds upon existing e-commerce and social media frameworks but tailors them specifically for martial artists. It extends the functionality of platforms like Shopify and Instagram by integrating martial arts-specific features such as belt tracking, training logs, and competition records.

1.3 Contributions

Key contributions of *Martial World* include:

- A dedicated e-commerce hub for martial arts gear and nutrition.
- A social media platform focused solely on martial arts content.
- A progress tracker for training, belt promotions, and competition history.
- Integration of Firebase for real-time updates and AWS for secure media storage.
- A scalable architecture using React Native and Django REST framework.

These contributions address the limitations of existing systems and offer a comprehensive solution tailored to the martial arts community.

1.4 Report Organization

- **Chapter 2**: System Study – Overview of existing systems and the proposed solution.
- **Chapter 3**: Methodology – Development approach, tools used, and module breakdown.
- **Chapter 4**: Results and Discussions – Screenshots and analysis of the implemented system.
- **Chapter 5**: Conclusion – Summary, achievements, and future scope.

Chapter 2. System Study

Martial World is designed to address the fragmented digital experience faced by martial artists. The system aims to unify commerce, community, and personal progress tracking into a single, discipline-specific platform.

2.1 Existing System

Currently, martial artists rely on multiple platforms to fulfill their needs—e-commerce sites for gear, social media for community engagement, and spreadsheets or journals for tracking progress. These systems are generic and not tailored to martial arts, leading to inefficiencies, lack of personalization, and diluted content. For example, platforms like Instagram or Amazon do not offer martial arts-specific filters or features. This fragmentation hinders growth and community cohesion.

Improving upon these limitations, *Martial World* proposes a unified solution that integrates all essential features into one seamless experience.

2.2 Proposed System

The proposed system, *Martial World*, is a mobile-first application that centralizes martial arts commerce, social networking, and progress tracking. It offers:

- A curated marketplace for gear, nutrition, and event tickets.
- A media-sharing platform focused solely on martial arts content.
- A personal dashboard for tracking training, belt promotions, and competition history.

Technologically, the system uses React Native for cross-platform development, Django REST for backend services, and Firebase for real-time updates. It is designed to be scalable, secure, and user-friendly.

Target users include martial artists of all levels, instructors, academies, and vendors specializing in martial arts products.

2.3 Functionalities of Proposed System

- **E-Commerce Hub:**
- Marketplace for gear, apparel, weapons, nutrition, and event tickets.
- Vendor onboarding and discipline-specific product categorization.
- Secure payment integration with Stripe and Razorpay.
- **Entertainment & Social Network:**
- Media sharing: photos, videos, reels, and stories.
- Follower system, hashtags, and content discovery.
- Personal branding tools for fighters, instructors, and academies.
- **Progress Tracker:**
- Training logs and belt promotion records.
- Goal-setting and fitness analytics.
- Journal entries and performance insights.

Chapter 3. Methodology

3.1 Introduction

To develop *Martial World*, an Agile methodology was adopted. Agile promotes iterative development, continuous feedback, and adaptive planning—ideal for building user-centric applications like this one. The project was divided into sprints, each focusing on specific modules such as e-commerce, social networking, and progress tracking. This approach ensured flexibility, faster delivery, and better alignment with user needs.

3.2 Software Tools

Operating System	Windows 11
Front End	JavaScript, HTML, CSS etc..
Back End	Firebase
Database	Firebase
IDE	Visual Studio Code
Version Control	Git

Table 3.1: List ofsoftware tools or languages used for the project development

3.3 Module Description

The system is divided into modular components to ensure maintainability and scalability.

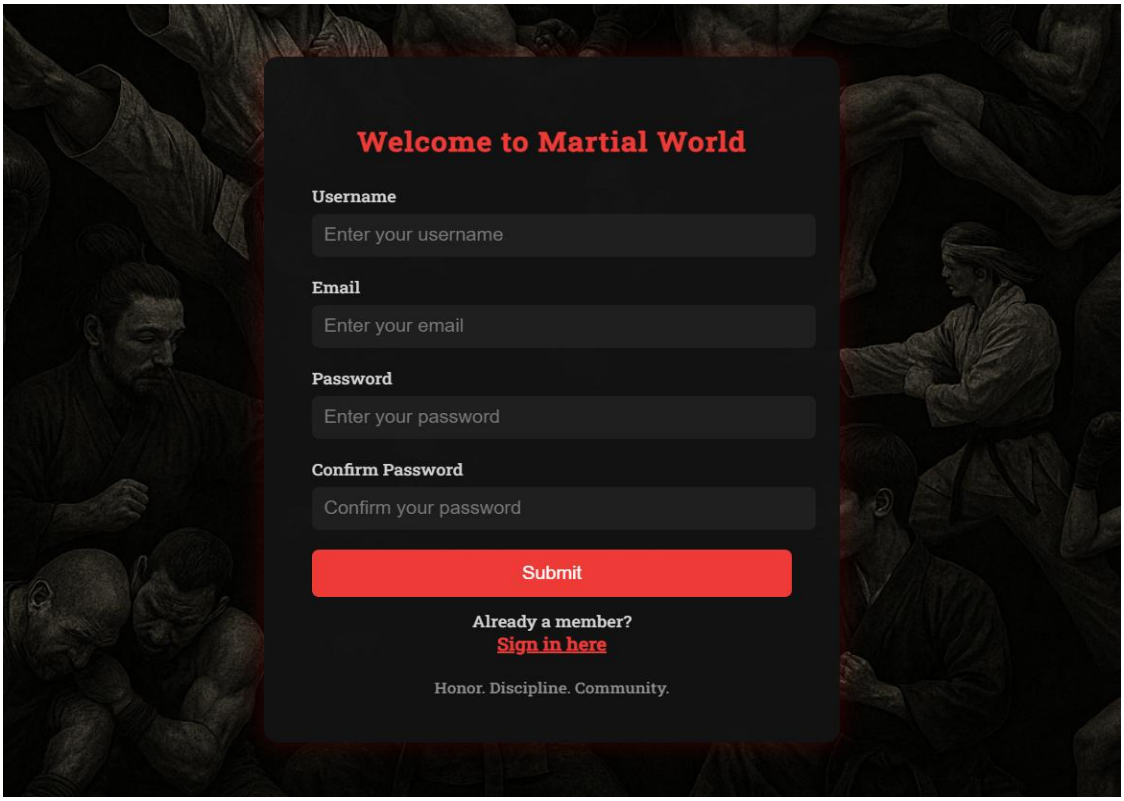
3.3.1 User Authentication Module

This model handles the signup an signin of the users. Ensures login for only registered users.

SIGNUP PAGE

```
<body>
  <div class="login-container">
    <h2>Welcome to Martial World</h2>
      <form id="loginForm">
        <label for="username">Username</label>
        <input type="text" id="username" placeholder="Enter your username" required />
        <label for="email">Email</label>
        <input type="email" id="email" placeholder="Enter your email" required />
        <label for="password">Password</label>
        <input type="password" id="password" placeholder="Enter your password" required />
        <label for="confirmPassword">Confirm Password</label>
        <input type="password" id="confirmPassword" placeholder="Confirm your password"
        required />
        <button type="submit">Submit</button>
        <div style="text-align: center; margin-top: 10px;">
          <span style="color: #ccc; font-size: 0.9rem;">Already a member?</span>
          <br />
          <a href="signin.html" style="color: #e53935; text-decoration: underline; font-weight:
          bold;">Sign in here</a>
        </div>
```

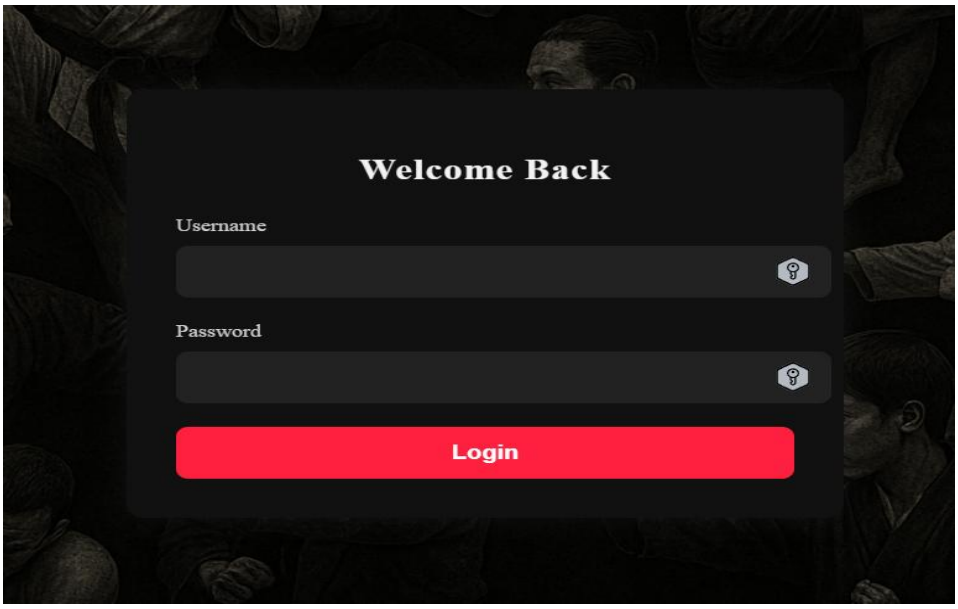
```
        </form>
        <div class="footer">Honor. Discipline. Community.</div>
    </div>
</body>
```



SIGNIN PAGE

```
<body>
    <div class="login-container">
        <h2>Welcome Back</h2>
        <form id="signinForm">
            <label for="username">Username</label>
```

```
    <input type="text" id="username" name="username" required />
    <label for="password">Password</label>
    <input type="password" id="password" name="password" required />
    <button type="submit">Login</button>
  </form>
 </div>
</body>
```



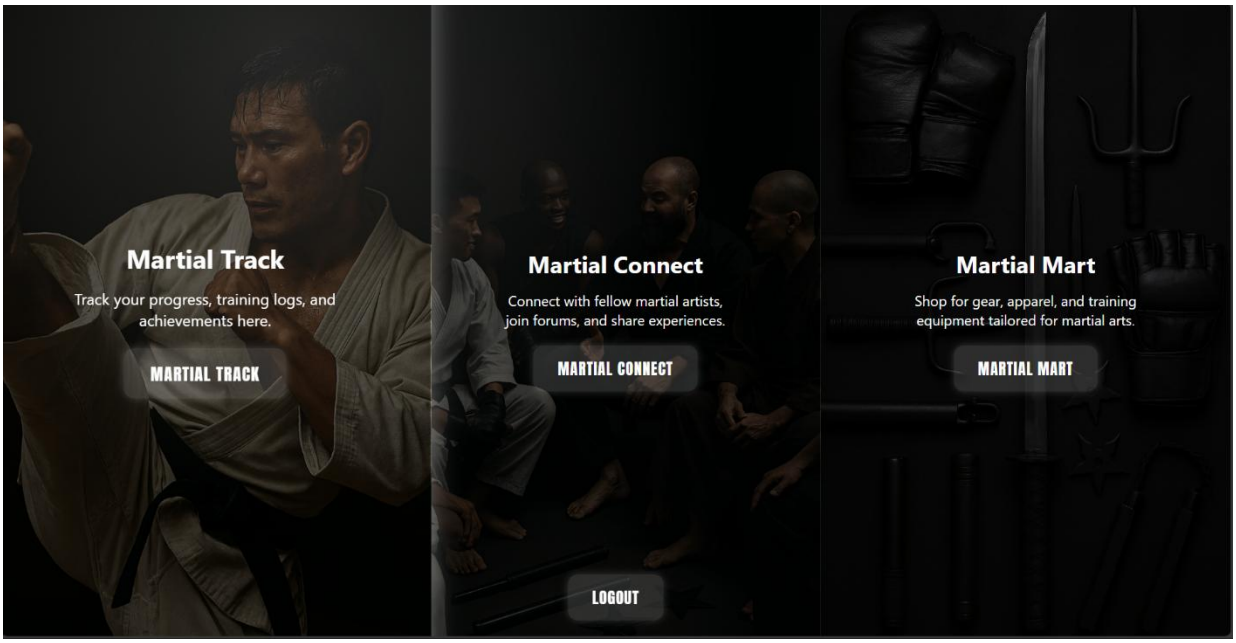
3.3.2 Progress Tracker Module

This module empowers users to log their martial arts journey. They can choose martial arts of their choice and can start their journey. Each can focus on their own martial arts and can train with the syllabus already provided in the page and also they can set timer for each section. They can also choose their level and start working according to their belt level.

TRACKER DASHBOARD

```
<body>
```

```
<div class="section track">
  <div class="section-content">
    <h2>Martial Track</h2>
    <p>Track your progress, training logs, and achievements here.</p>
    <button onclick="location.href='martial-track.html'">Martial Track</button>
  </div>
</div>
```

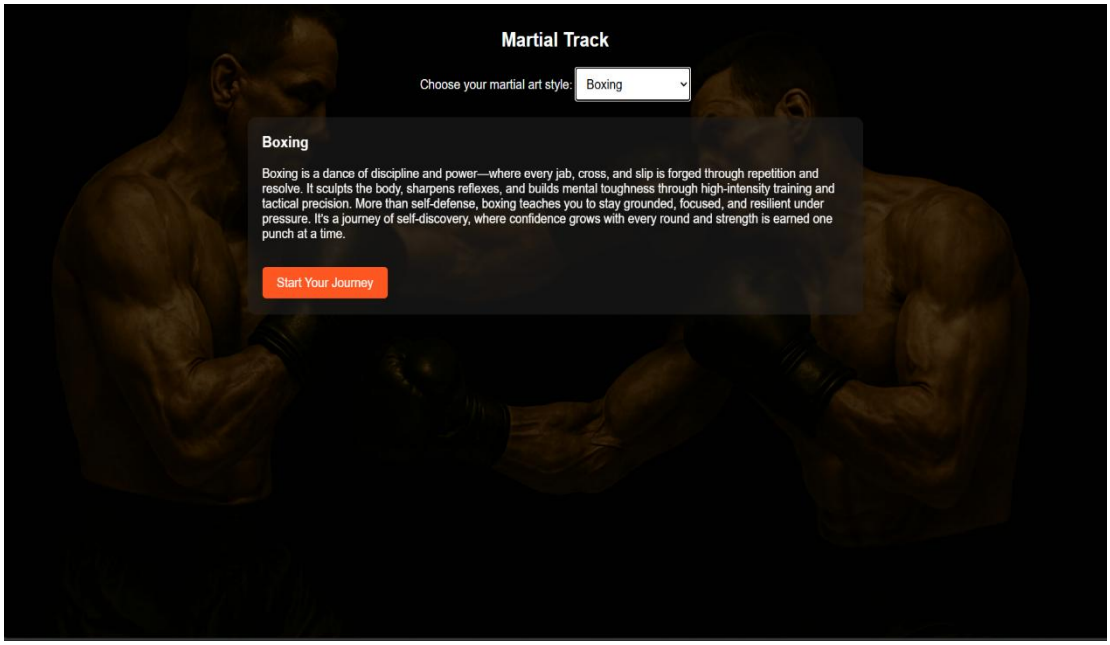


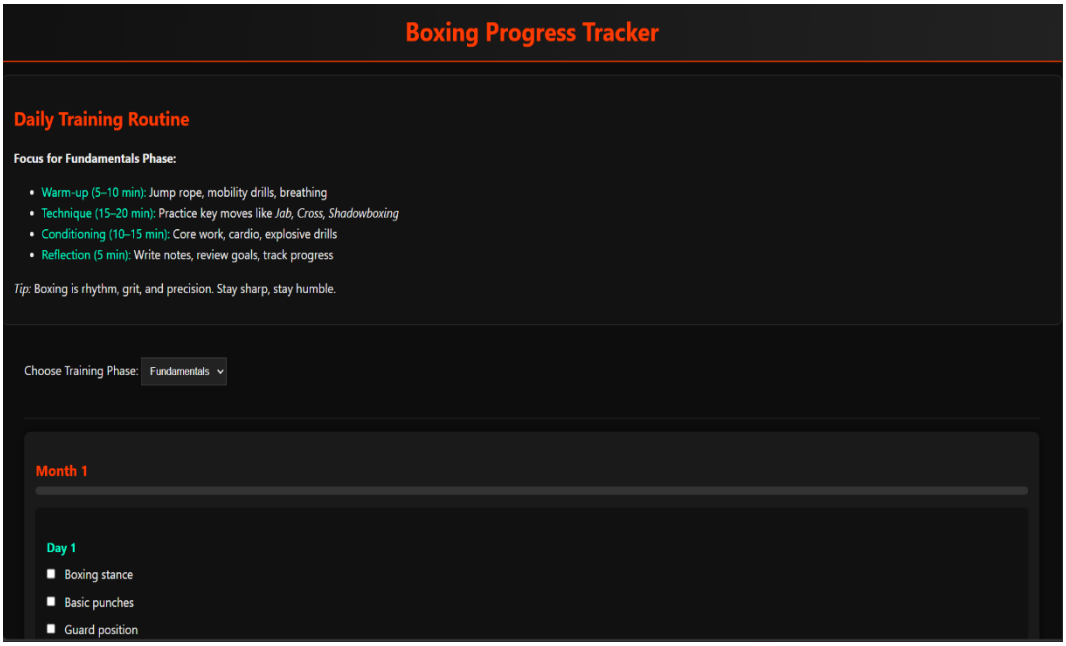
TRACKER MENU

```
<body>
  <div class="overlay">
    <h2>Martial Track</h2>


    <form id="styleForm">
```

```
      <label for="martialArt">Choose your martial art style:</label>
      <select id="martialArt" required>
        <option value="">-- Select Style --</option>
        <option value="Kungfu">Kungfu</option>
        <option value="Karate">Karate</option>
        <option value="Taekwondo">Taekwondo</option>
        <option value="Muay Thai">Muay Thai</option>
        <option value="Boxing">Boxing</option>
        <option value="Kickboxing">Kickboxing</option>
        <option value="MMA">MMA</option>
      </select>
    </form>

    <div id="descriptionBox">
      <h3 id="styleTitle"></h3>
      <p id="styleDescription"></p>
      <button id="nextButton">Start Your Journey</buton>
    </div>
  </div>
</body>
<script>
```

```
const martialDescriptions = {
  "Kungfu": {
    text: "Kung Fu is more than combat…",
    image: "url('https://copilot.microsoft.com/th/id/BCO.13e8b530-70b0-4368-a57c-94c820df3804.png')",
    link: "kungfu.html"
.
.
.
  },
};
```



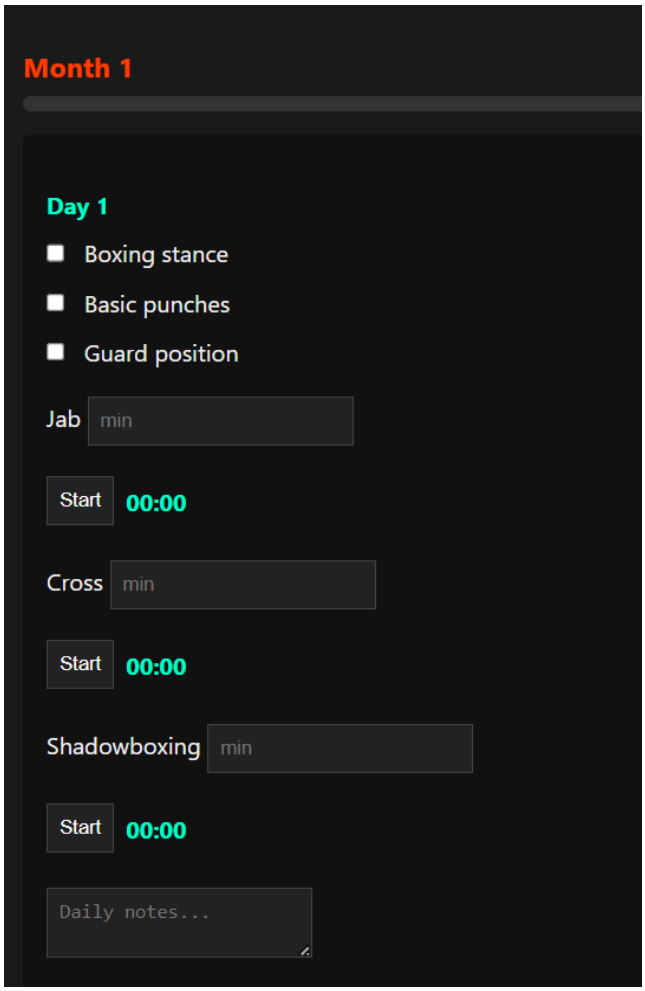


TIMER

```
function startTimer(button) {

    const exerciseDiv = button.closest(".exercise");

    const input = exerciseDiv.querySelector(".timeInput");

    const display = exerciseDiv.querySelector(".timer-display");


    let minutes = parseInt(input.value);

    if (isNaN(minutes) || minutes <= 0) {

      alert("Enter a valid number of minutes.");

      return;

    }


    let seconds = minutes * 60;
```

```
  button.disabled = true;
  display.textContent = formatTime(seconds);

  const interval = setInterval(() => {
    seconds--;
    display.textContent = formatTime(seconds);

    if (seconds <= 0) {
      clearInterval(interval);
      button.disabled = false;
      display.textContent = "Time's up!";
      playAlarm();
      exerciseDiv.style.border = "2px solid #ff3c00";
    }
  }, 1000);
}

function formatTime(sec) {
  const m = Math.floor(sec / 60);
  const s = sec % 60;
  return `${m.toString().padStart(2, '0')}:${s.toString().padStart(2, '0')}`;
```

```
}

function playAlarm() {
  const audio = new Audio("https://actions.google.com/sounds/v1/alarms/alarm_clock.ogg");
  audio.play();
}
```



3.4 User Story

Role	Goal	Benefit
Martial Artist	Buy gear and track progress	Saves time and improves focus
Instructor	Share content and manage students	Builds reputation and engagement
Vendor	Sell martial arts products	Reaches targeted audience

3.5 Product Backlog

ID	Feature	Priority	Status
1	User Registration/Login	High	Completed
2	Progress Tracker	High	Completed
3	Media Sharing	Medium	In Progress
4	Product Listing	Low	Planned
5	Journal Entries	Low	Planned

3.6 Project Plan

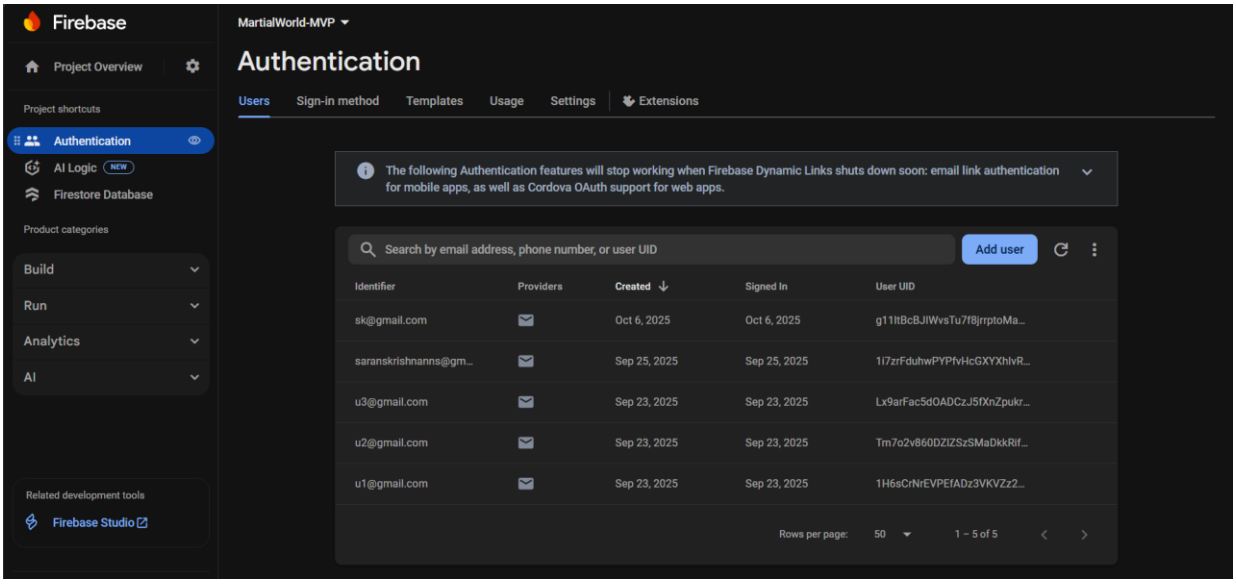
Sprint	Duration	Focus Area
1	2 weeks	Authentication & UI
2	2 weeks	Progress Tracker
3	2 weeks	Social Features

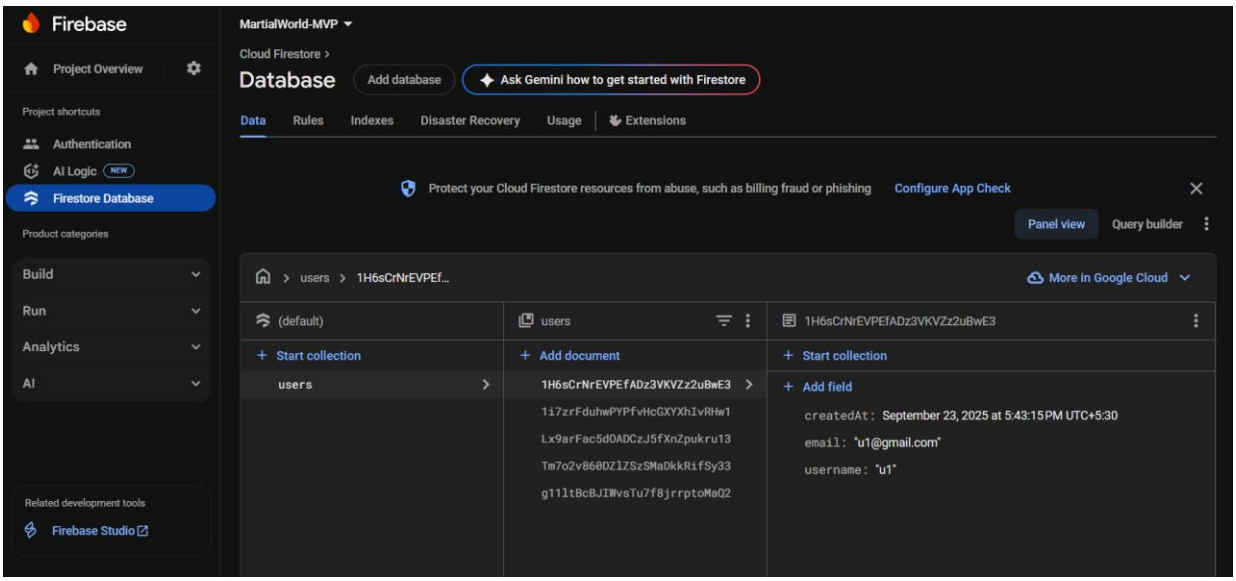
3.7 Sprint Backlog

Sprint	Task	Status
1	Firebase Auth Setup	Completed
1	UI Design for Login/Register	Completed
2	Product API Integration	Completed
3	Media Upload Feature	In Progress

3.8 Database Design

The database schema includes collections of User details like Username, Email address.





Chapter 4. Results and Discussions

4.1 Results

The *Martial World* application was successfully developed and tested across its core modules. Below are key screenshots and descriptions of the implemented features:

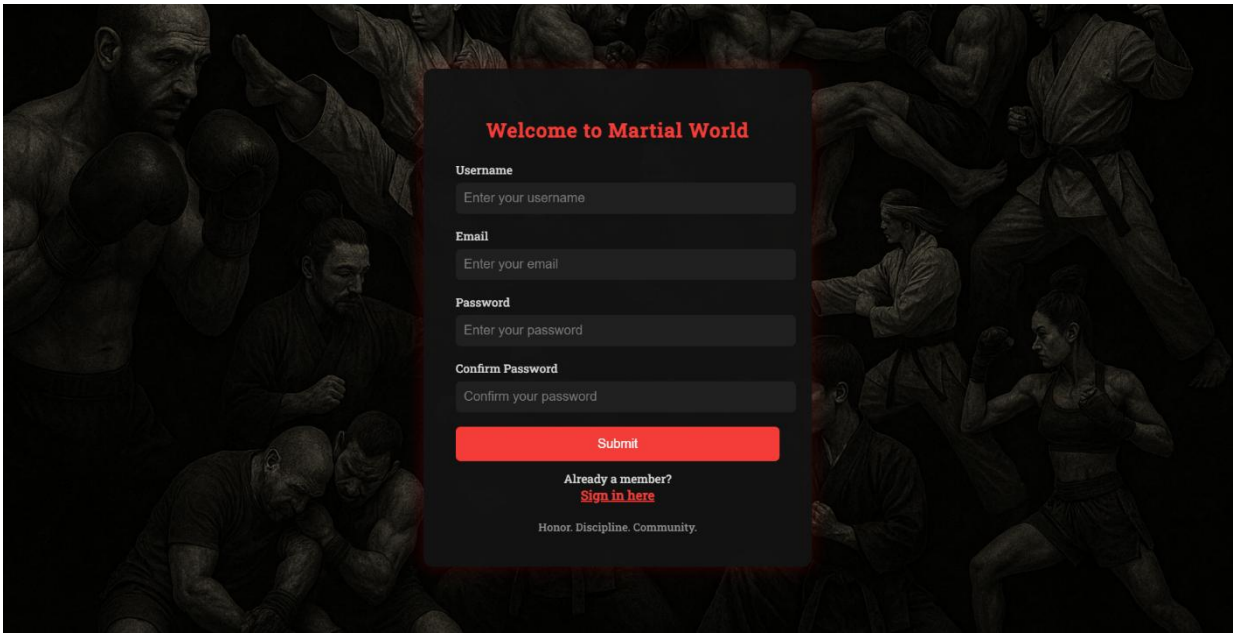


Figure 4.1: SignUp Page

This is the welcome screen of the app, featuring a clean UI with options to log in, register. It sets the tone for a martial arts-focused experience.

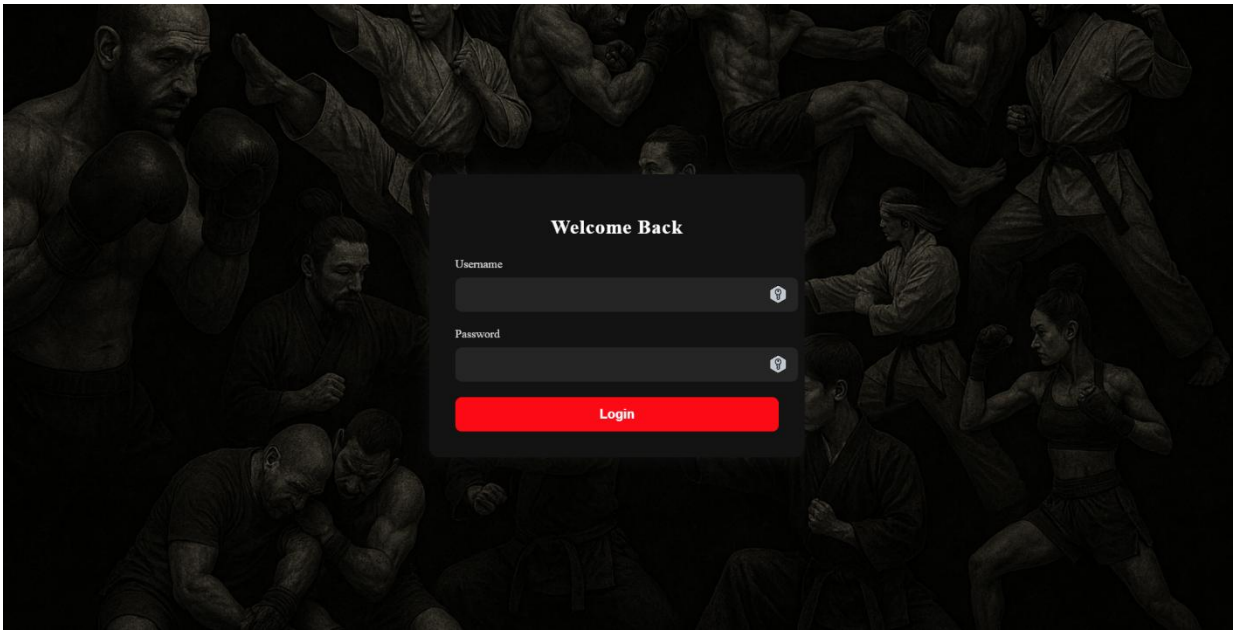


Figure 4.2: SignIn Page

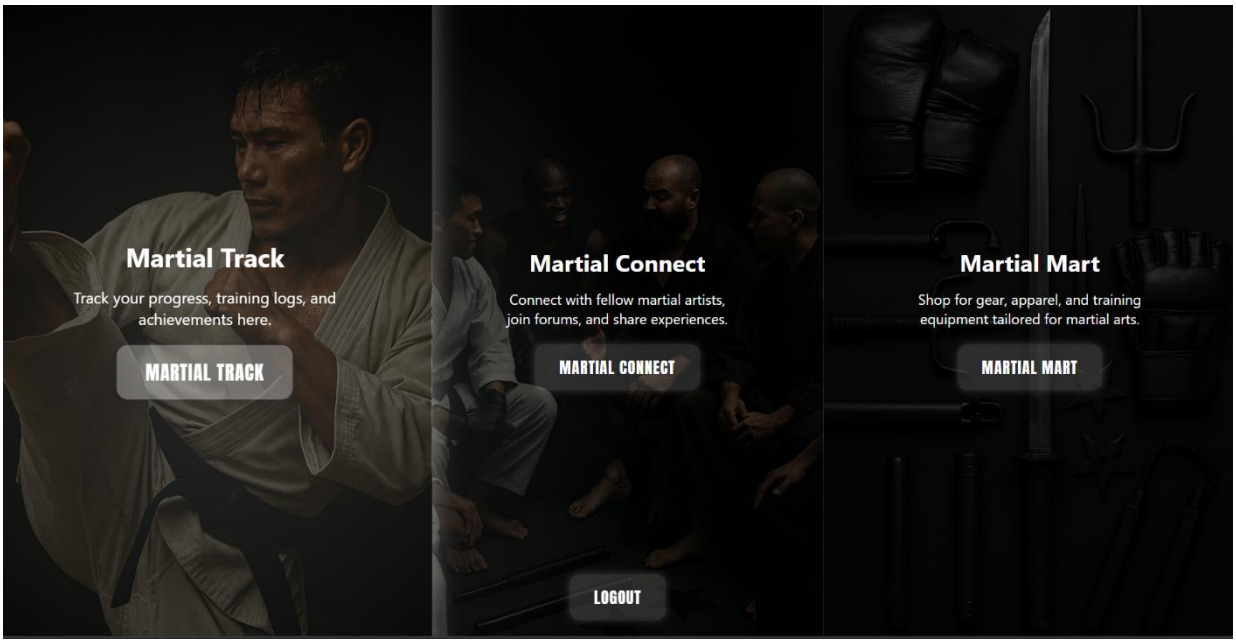


Figure 4.3: Dashboard Page

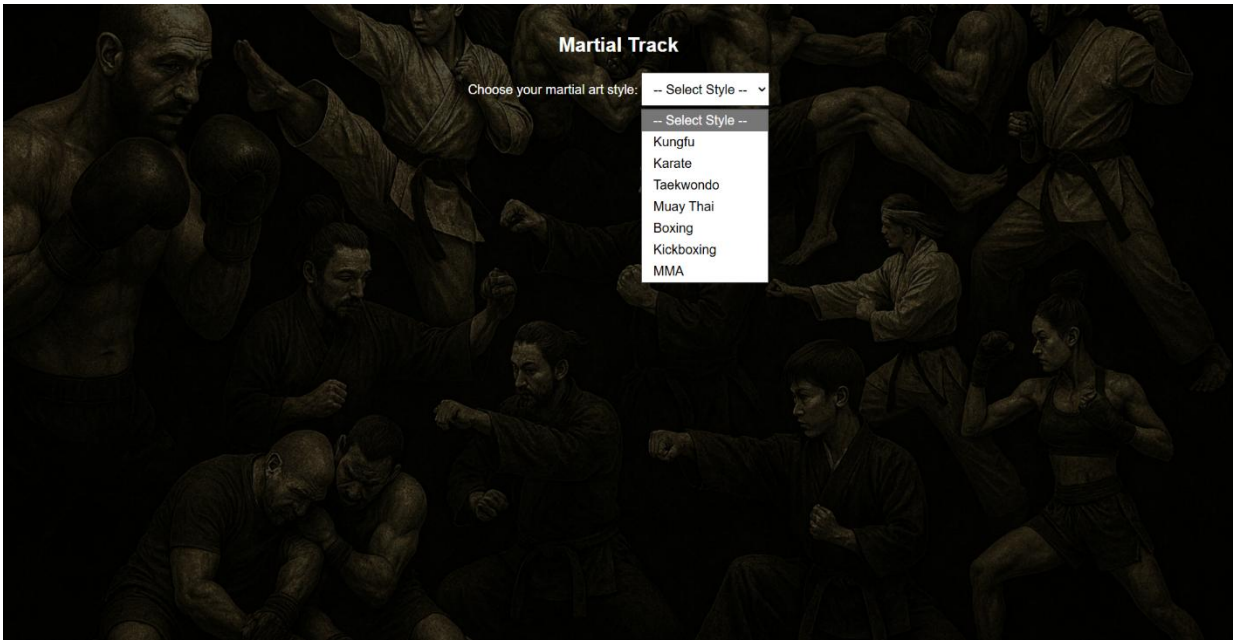


Figure 4.4: Martial Track Page

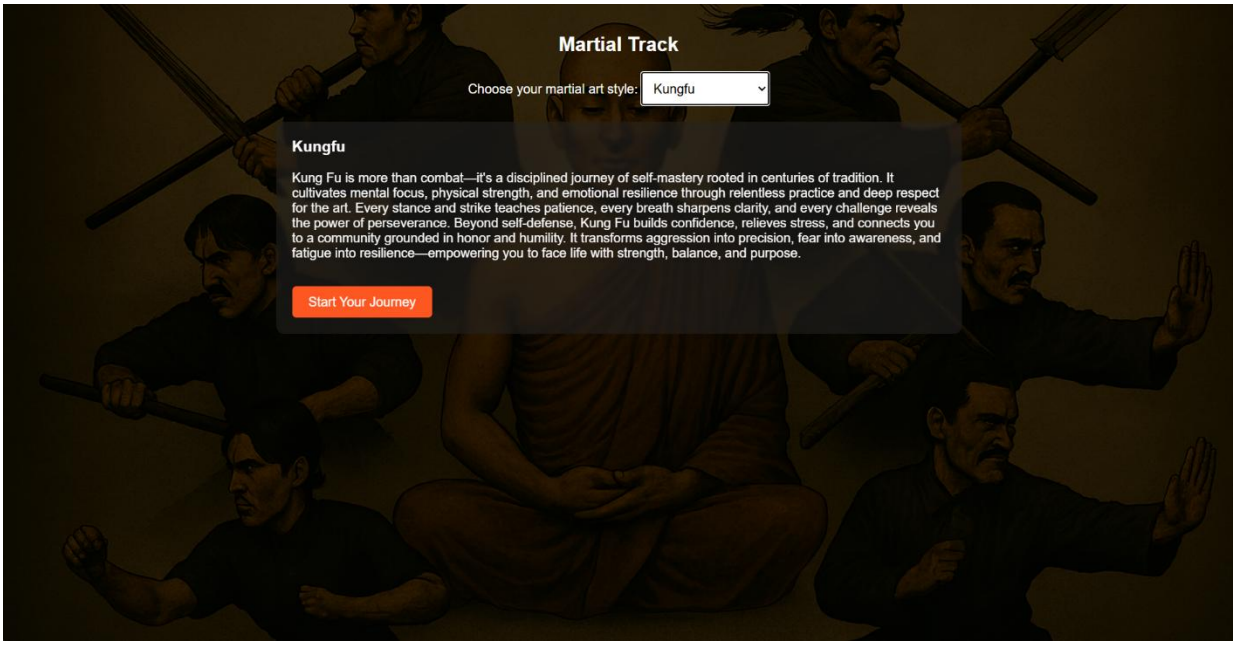


Figure 4.5: Kungfu Page

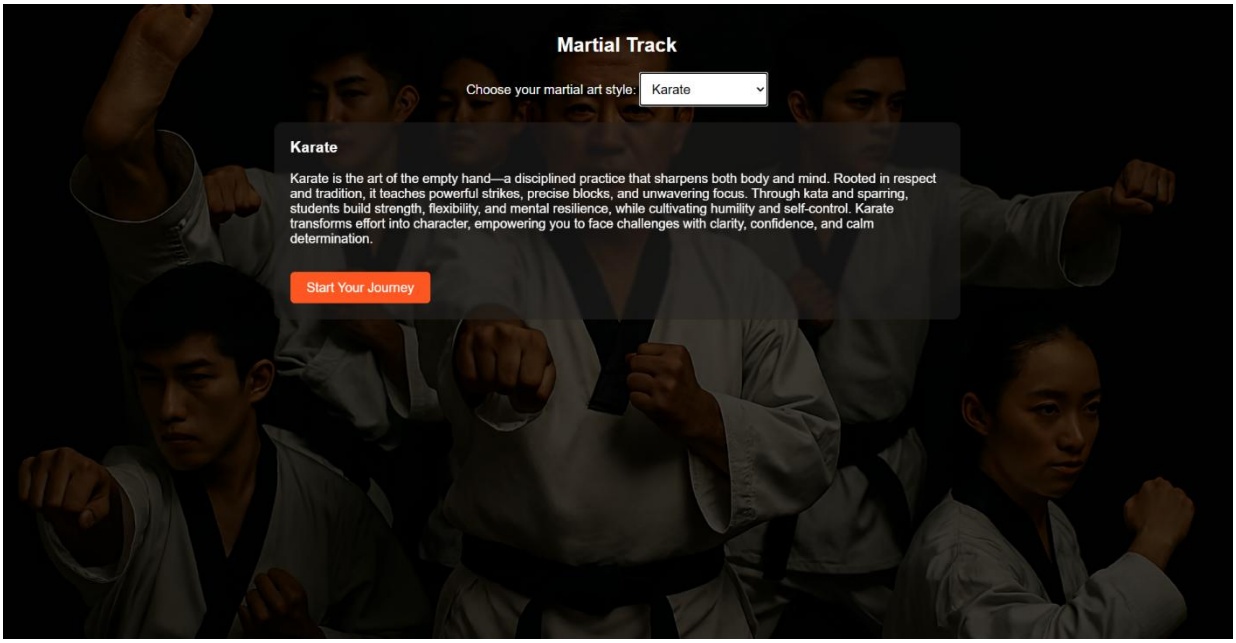


Figure 4.6: Karate Page

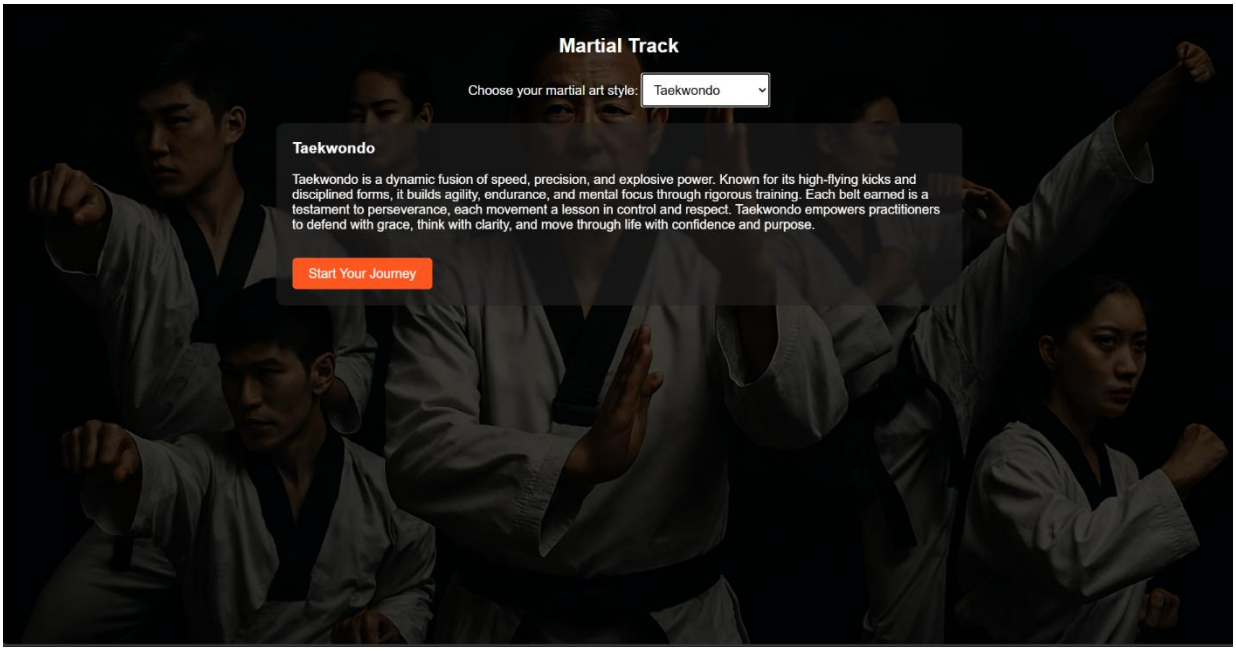


Figure 4.7: Taekwondo Page

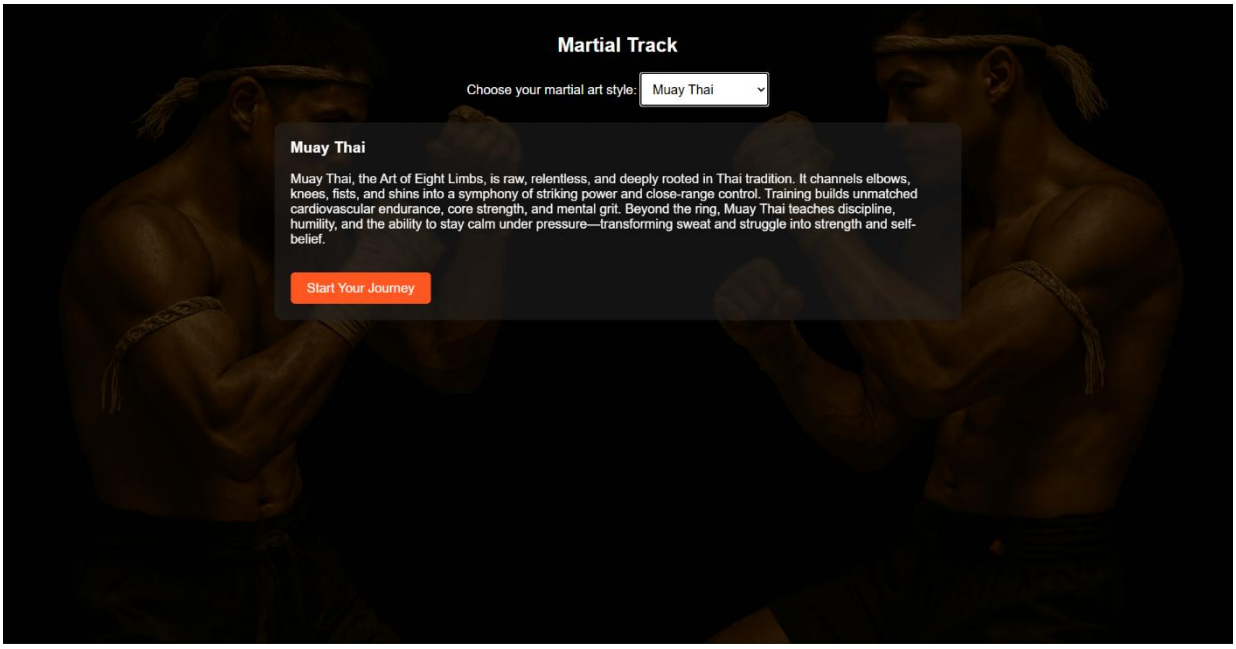


Figure 4.8: Muay Thai Page

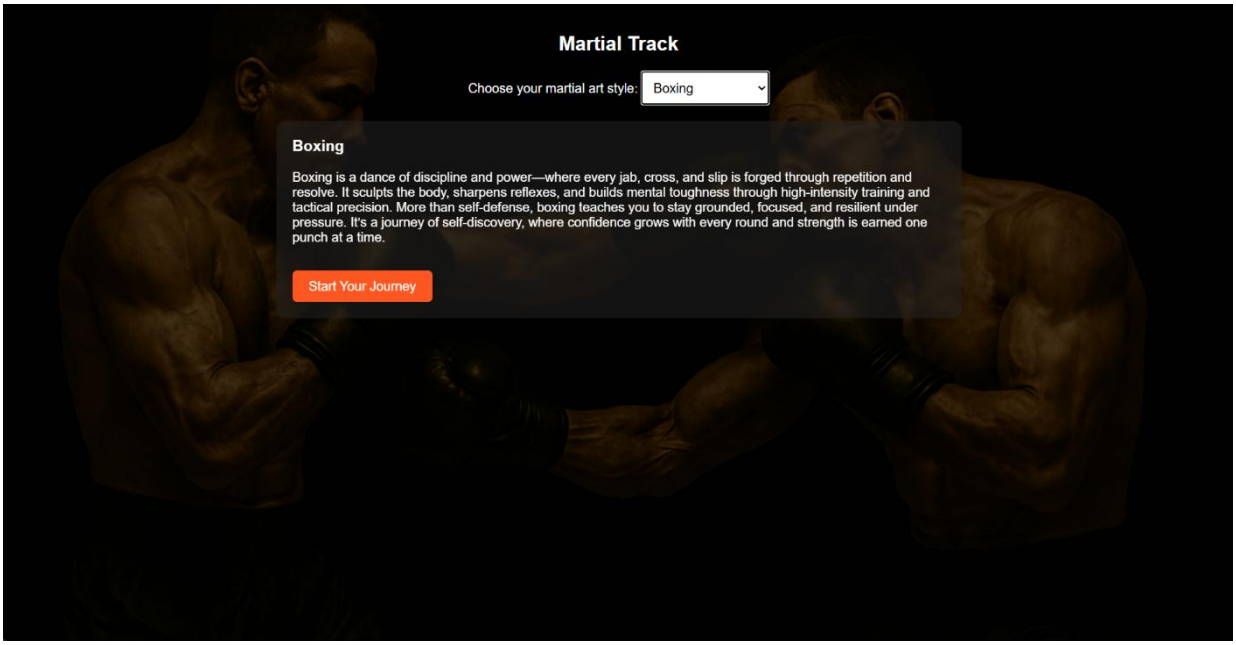


Figure 4.9: Boxing Page

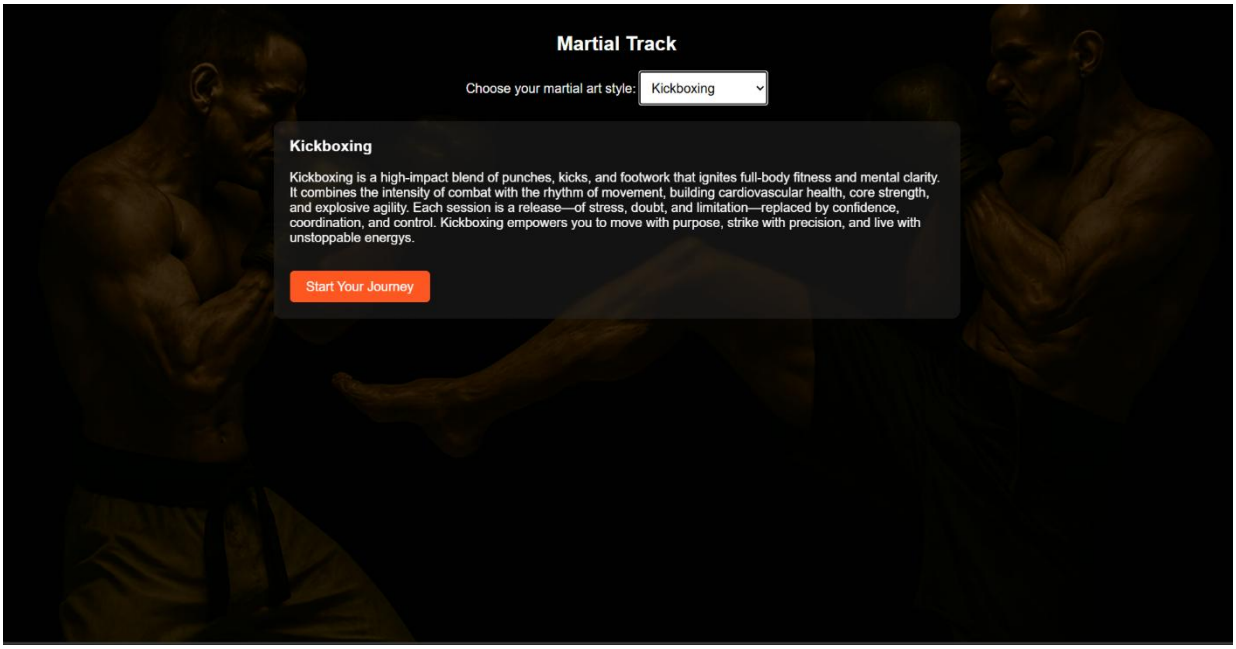


Figure 4.10: Kickboxing Page

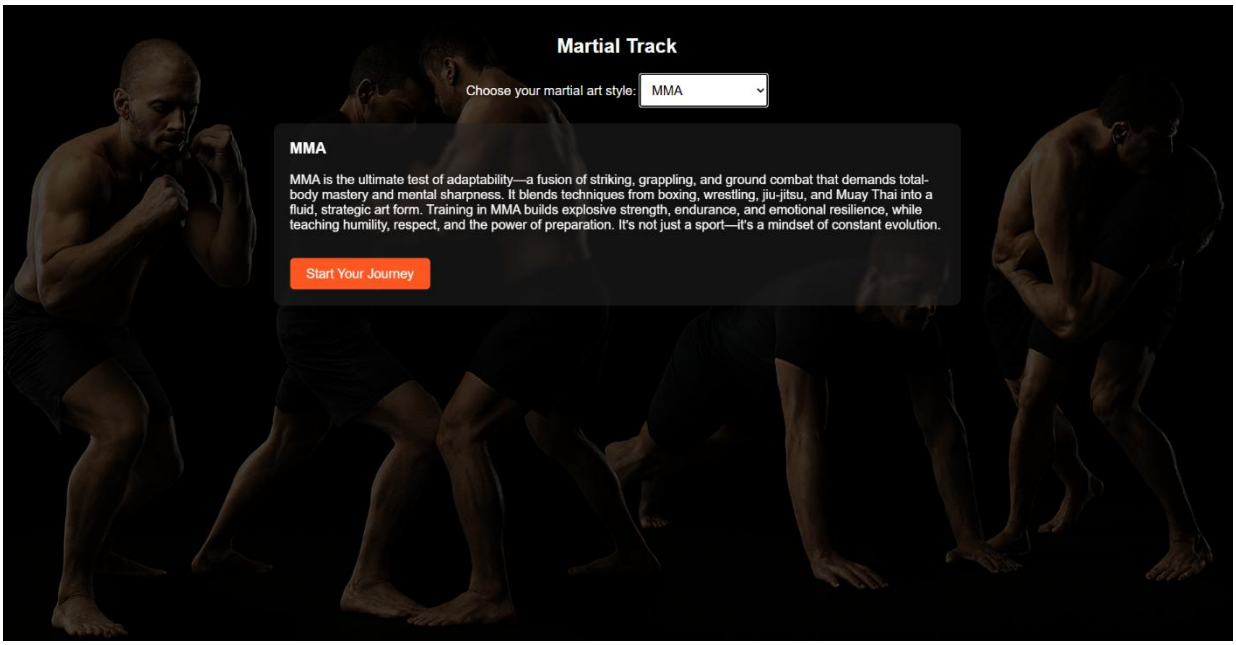


Figure 4.11: MMA Page

Figure 4.12: Kungfu Tracking Page

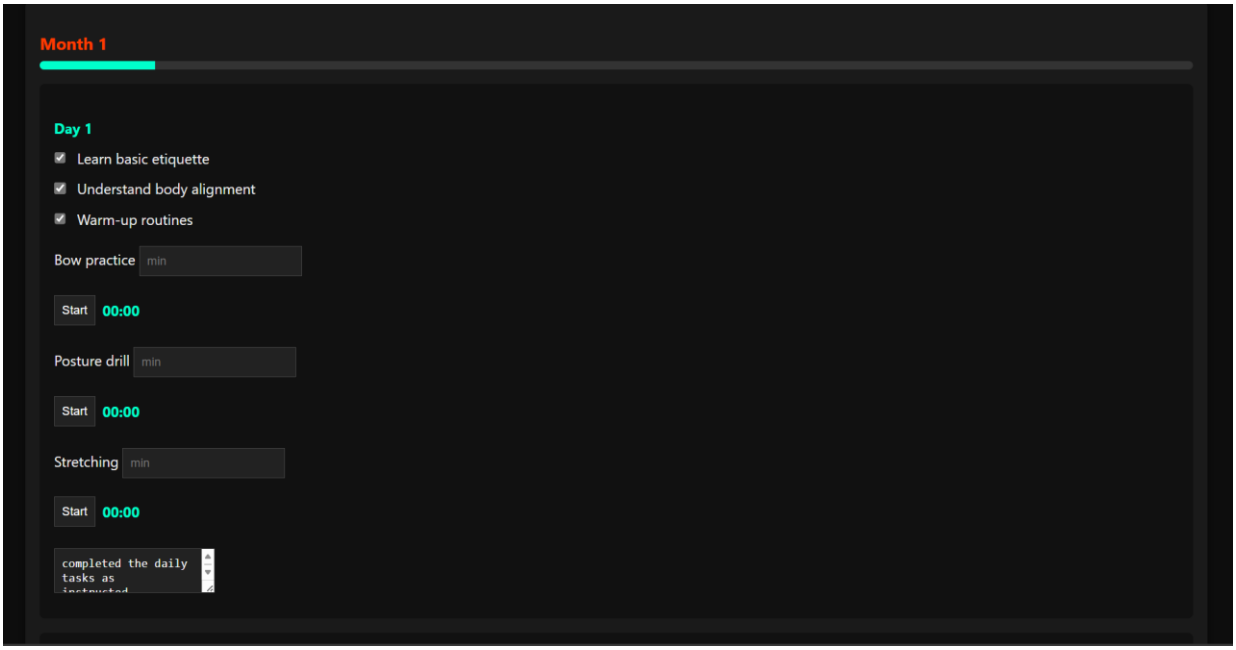


Figure 4.12.1: Kungfu Tracking Page

Chapter 5. Conclusion

Martial World is a comprehensive web application designed to unify the martial arts community through a dedicated digital ecosystem. It successfully integrates commerce, social networking, and personal development tools tailored to martial artists.

Key achievements include:

- A discipline-specific marketplace for gear and nutrition.
- A vibrant social platform for sharing martial arts content.
- A robust progress tracker for training, belt promotions.

The project demonstrates the power of full-stack development in solving real-world problems. While the MVP covers core functionalities, future enhancements could include creating own syllabus and academy management tools.

Limitations faced included time constraints for advanced features like live streaming and vendor analytics. These can be addressed in future iterations with extended development cycles and user feedback.

References

[1] Firebase, “Firebase Documentation,” Google LLC, [Online]. Available: https://firebase.google.com/docs. . [Accessed: 01 Oct 2025].

[2] WAKO India Kickboxing Federation, “National Kickboxing Guidelines,” [Online]. Available: https://wakoindia.org/. . [Accessed: 01 Oct 2025].

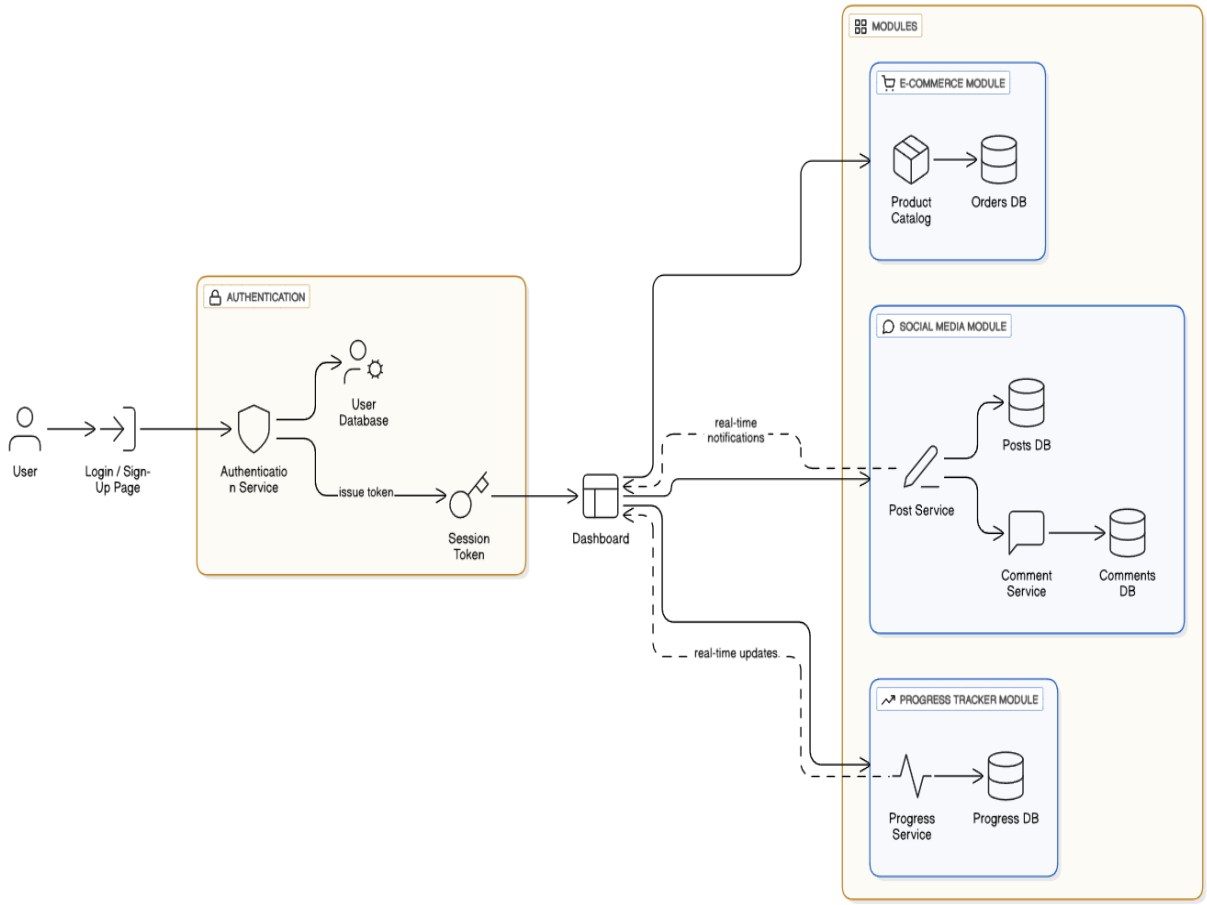
[3] "Getting started with Firebase on the Web" A concise walkthrough of setting up Firebase, including authentication, Firestore, and hosting. Ideal for web-based apps.

[4] HTML, CSS, and Javascript in 30 minutes A fast-paced crash course that introduces all three technologies in a single session—great for quick revision.

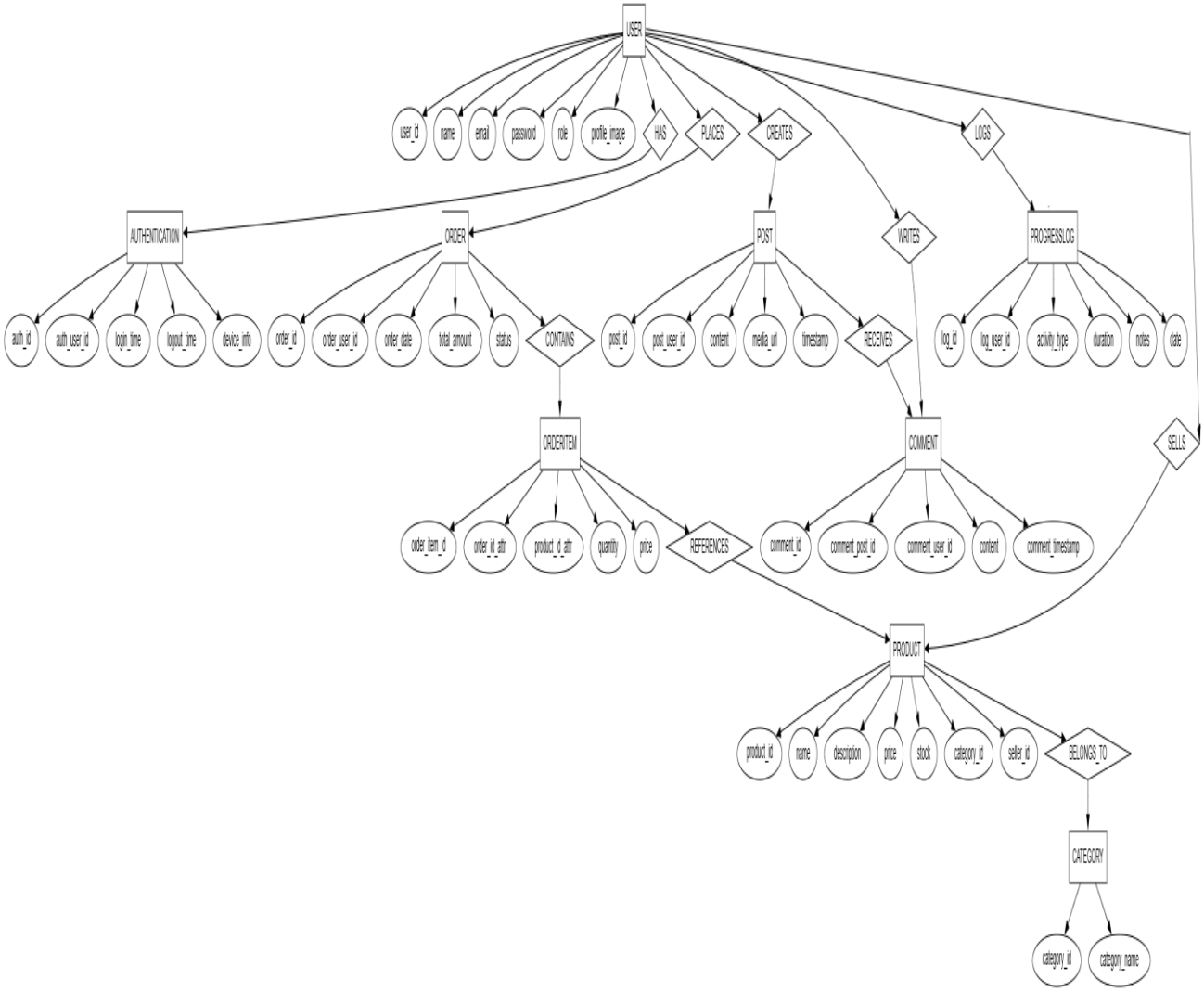
[5] Microsoft Copilot, AI Companion for Technical and Creative Assistance. [Online]. Available: https://copilot.microsoft.com. [Accessed: 06 Oct 2025].

Appendix

Appendix A Data Flow Diagram



Appendix B ER Diagram



Appendix C Source Code

1. Sign Up Page

```html
<body>
  <div class="login-container">
    <h2>Welcome to Martial World</h2>
    <form id="loginForm">
      <label for="username">Username</label>
      <input type="text" id="username" placeholder="Enter your username" required />

      <label for="email">Email</label>
      <input type="email" id="email" placeholder="Enter your email" required />

      <label for="password">Password</label>
      <input type="password" id="password" placeholder="Enter your password" required />

      <label for="confirmPassword">Confirm Password</label>
      <input type="password" id="confirmPassword" placeholder="Confirm your password" required />

      <button type="submit">Submit</button>
      <div style="text-align: center; margin-top: 10px;">
        <span style="color: #ccc; font-size: 0.9rem;">Already a member?</span>
        <br />
        <a href="signin.html" style="color: #e53935; text-decoration: underline; font-weight: bold;">Sign in here</a>
      </div>

    </form>
    <div class="footer">Honor. Discipline. Community.</div>
  </div>
```

2. Sign In Page

```html
<body>
  <div class="login-container">
    <h2>Welcome Back</h2>
    <form id="signinForm">
      <label for="username">Username</label>
      <input type="text" id="username" name="username" required />

      <label for="password">Password</label>
      <input type="password" id="password" name="password" required />

      <button type="submit">Login</button>
    </form>
  </div>
```

3. Dashboard

```html
<body>
  <div class="section track">
    <div class="section-content">
      <h2>Martial Track</h2>
      <p>Track your progress, training logs, and achievements here.</p>
      <button onclick="location.href='martial-track.html'">Martial Track</button>
    </div>
  </div>

  <div class="section connect">
    <div class="section-content">
      <h2>Martial Connect</h2>
      <p>Connect with fellow martial artists, join forums, and share experiences.</p>
      <button onclick="location.href='martial-connect.html'">Martial Connect</button>
    </div>
  </div>

  <div class="section mart">
    <div class="section-content">
      <h2>Martial Mart</h2>
      <p>Shop for gear, apparel, and training equipment tailored for martial arts.</p>
      <button onclick="location.href='martial-mart.html'">Martial Mart</button>
    </div>
  </div>

  <div id="mouse-light"></div>

  <div class="logout-container">
  <button onclick="logout()">Logout</button>
  </div>
```

4. Martial Track Menu

```html
<body>
  <div class="overlay">
    <h2>Martial Track</h2>

    <form id="styleForm">
      <label for="martialArt">Choose your martial art style:</label>
      <select id="martialArt" required>
        <option value="">-- Select Style --</option>
        <option value="Kungfu">Kungfu</option>
        <option value="Karate">Karate</option>
        <option value="Taekwondo">Taekwondo</option>
        <option value="Muay Thai">Muay Thai</option>
        <option value="Boxing">Boxing</option>
        <option value="Kickboxing">Kickboxing</option>
        <option value="MMA">MMA</option>
      </select>
    </form>

    <div id="descriptionBox">
      <h3 id="styleTitle"></h3>
      <p id="styleDescription"></p>
      <button id="nextButton">Start Your Journey</buton>
    </div>
  </div>
```

5. Progress Tracker Bar

```javascript
const progressPercent = totalGoals > 0 ? Math.round((completedGoals / totalGoals) * 100) : 0;
const progressFill = monthDiv.querySelector(`#progress-${belt}-${monthIndex}`);
if (progressFill) {
  progressFill.style.width = `${progressPercent}%`;
}
```

6. Timer

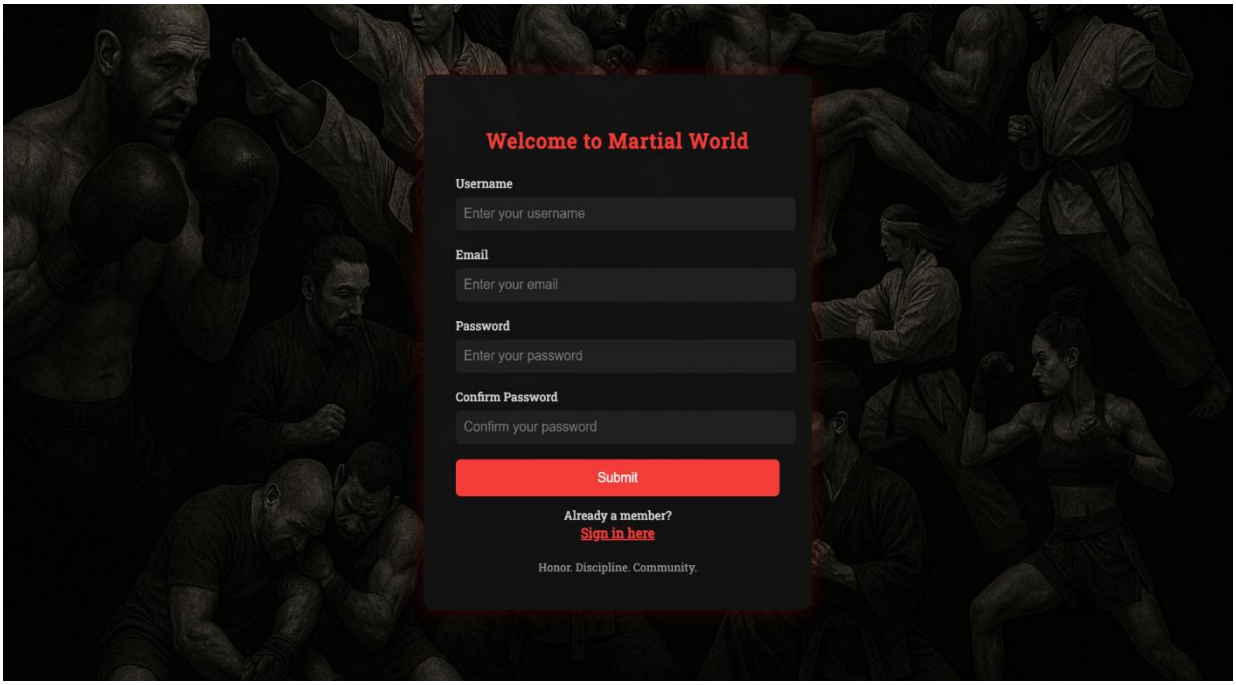
```
function startTimer(button) {
  const exerciseDiv = button.closest(".exercise");
  const input = exerciseDiv.querySelector(".timeInput");
  const display = exerciseDiv.querySelector(".timer-display");

  let minutes = parseInt(input.value);
  if (isNaN(minutes) || minutes <= 0) {
    alert("Enter a valid number of minutes.");
    return;
  }

  let seconds = minutes * 60;
  button.disabled = true;
  display.textContent = formatTime(seconds);

  const interval = setInterval(() => {
    seconds--;
    display.textContent = formatTime(seconds);

    if (seconds <= 0) {
      clearInterval(interval);
      button.disabled = false;
      display.textContent = "Time's up!";
      playAlarm();
      exerciseDiv.style.border = "2px solid #ff3c00";
    }
  }, 1000);
}

function formatTime(sec) {
  const m = Math.floor(sec / 60);
  const s = sec % 60;
  return `${m.toString().padStart(2, '0')}:${s.toString().padStart(2, '0')}`;
}

function playAlarm() {
  const audio = new Audio("https://actions.google.com/sounds/v1/alarms/alarm_clock.ogg");
  audio.play();
}

updateSyllabus(); // Load default on page load
```

7. Choosing Level

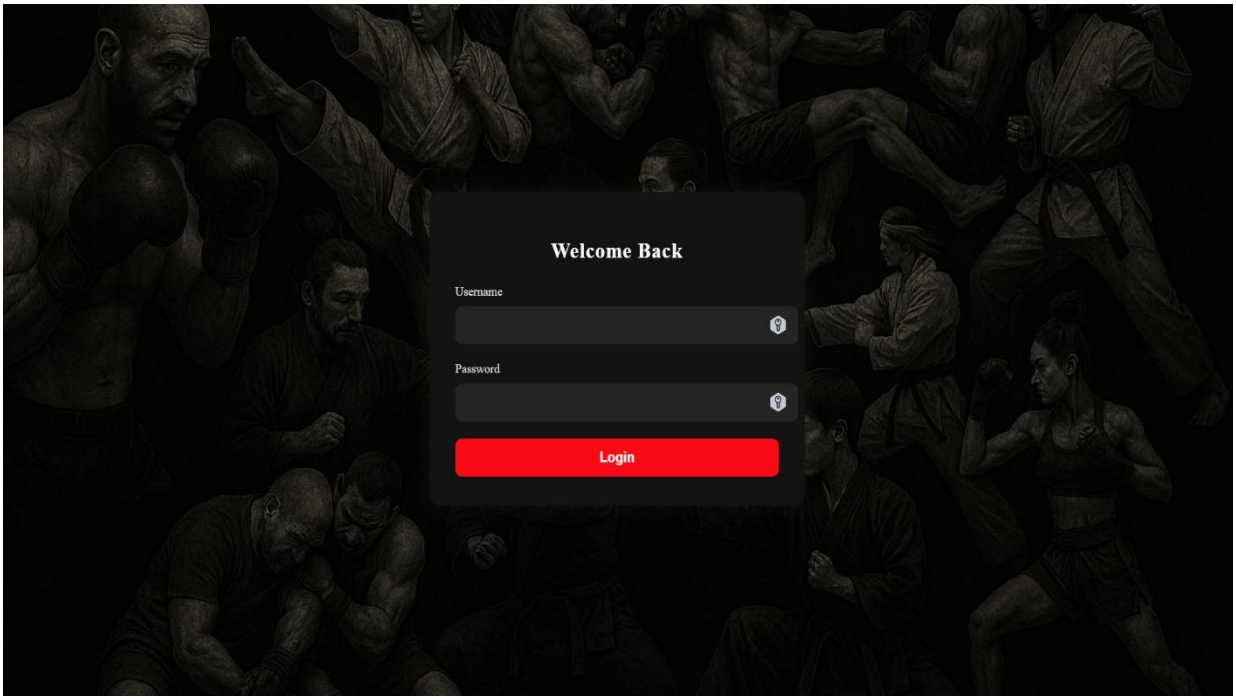
```
function updateSyllabus() {
  const belt = document.getElementById("belt").value;
  const content = document.getElementById("beltContent");
  const guide = document.getElementById("syllabusText");
  const data = syllabusData[belt];

  guide.innerHTML = `
    <strong>Focus for ${belt.charAt(0).toUpperCase() + belt.slice(1)} Belt:</strong><br/>
    <ul>
      <li><span style="color:#00ffcc;">Warm-up (5–10 min):</span> Stretching, breathing, light cardio</li>
      <li><span style="color:#00ffcc;">Technique (15–20 min):</span> Practice key moves like <em>${data.exercises.join(', ')}</em></li>
      <li><span style="color:#00ffcc;">Conditioning (10–15 min):</span> Strength drills, endurance work</li>
      <li><span style="color:#00ffcc;">Reflection (5 min):</span> Write notes, review goals, track progress</li>
    </ul>
    <em>Tip:</em> Stay consistent. Mastery comes from repetition, discipline, and honest effort.
  `;
```

Appendix D Screenshot

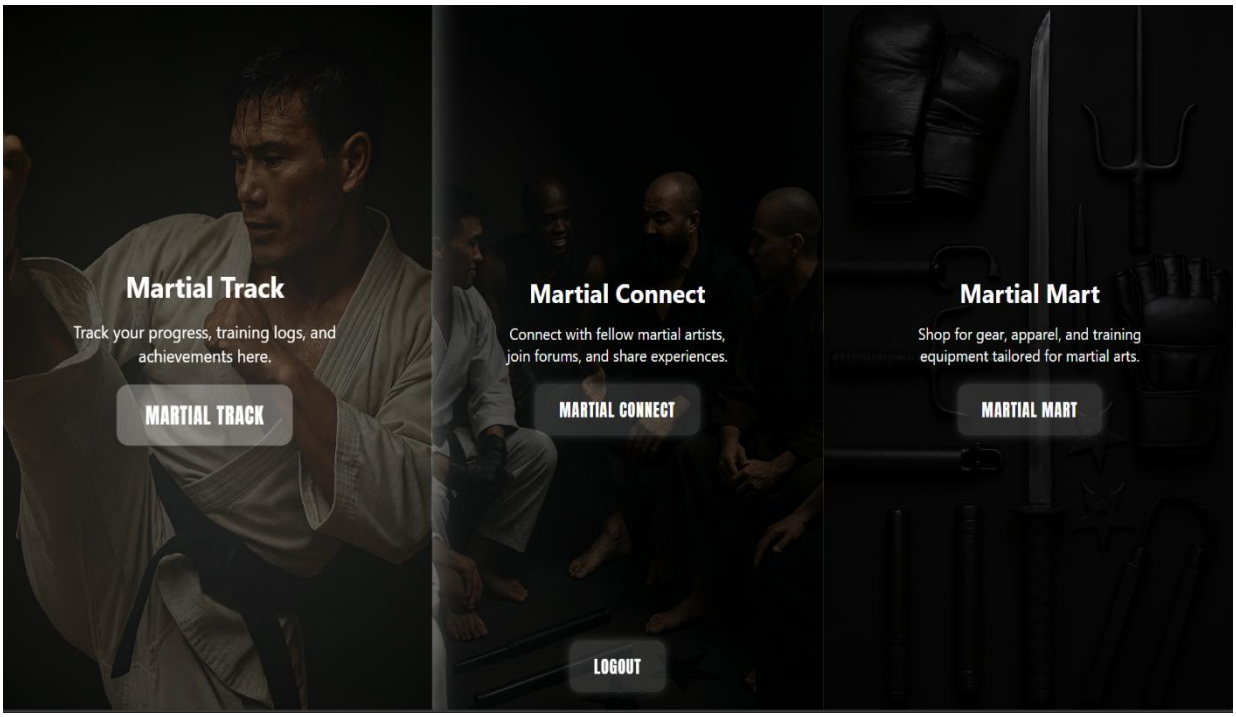
1. Sign Up Page



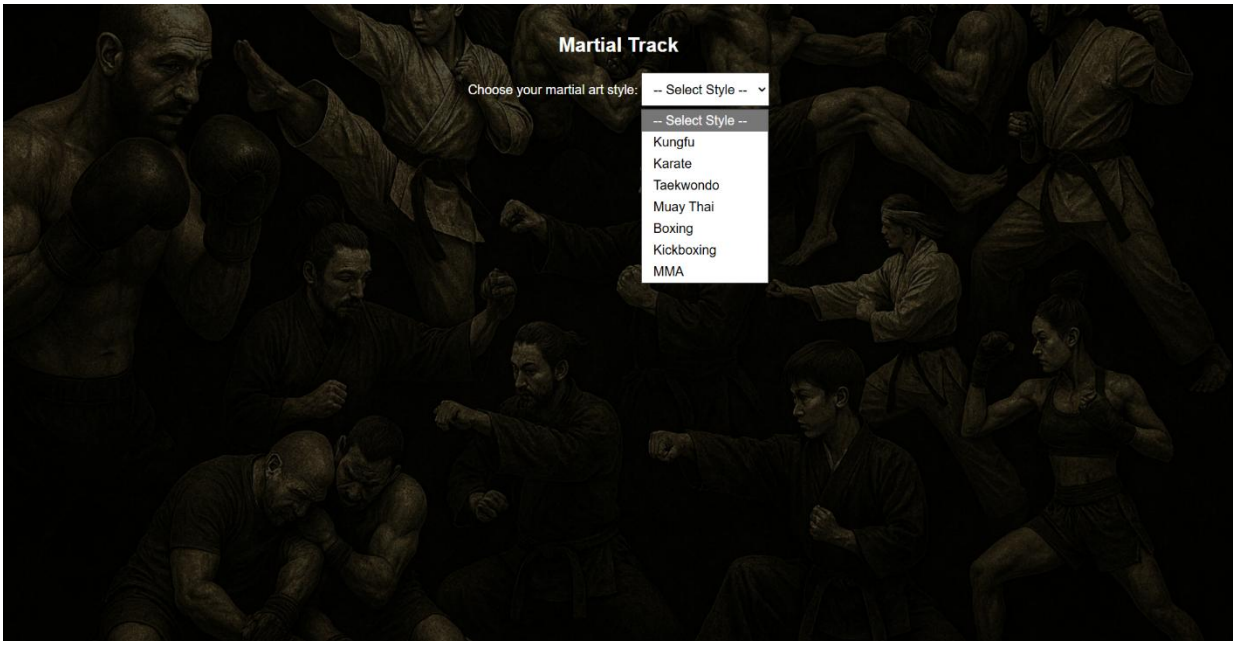
2. Sign In Page


Welcome Back
Username
Password
Login

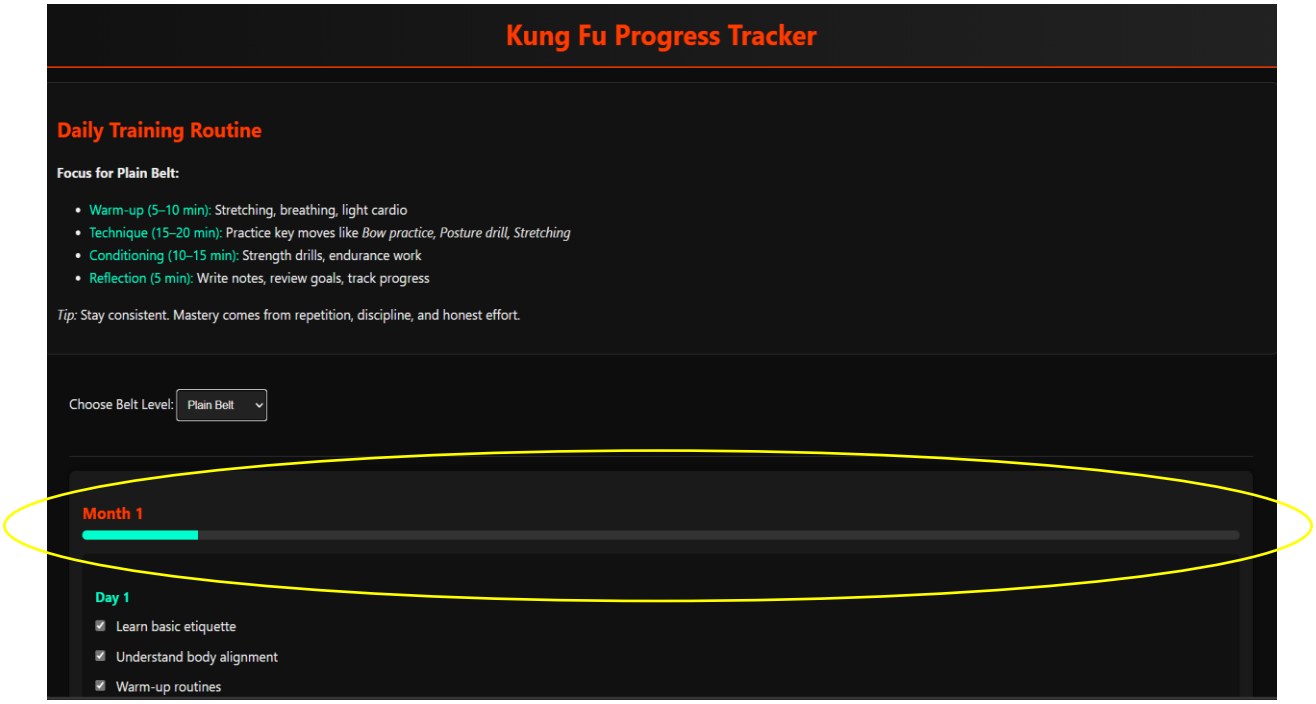
3. Dashboard


Martial Track
Track your progress, training logs, and achievements here.
MARTIAL TRACK
Martial Connect
Connect with fellow martial artists, join forums, and share experiences.
MARTIAL CONNECT
Martial Mart
Shop for gear, apparel, and training equipment tailored for martial arts.
MARTIAL MART
LOGOUT

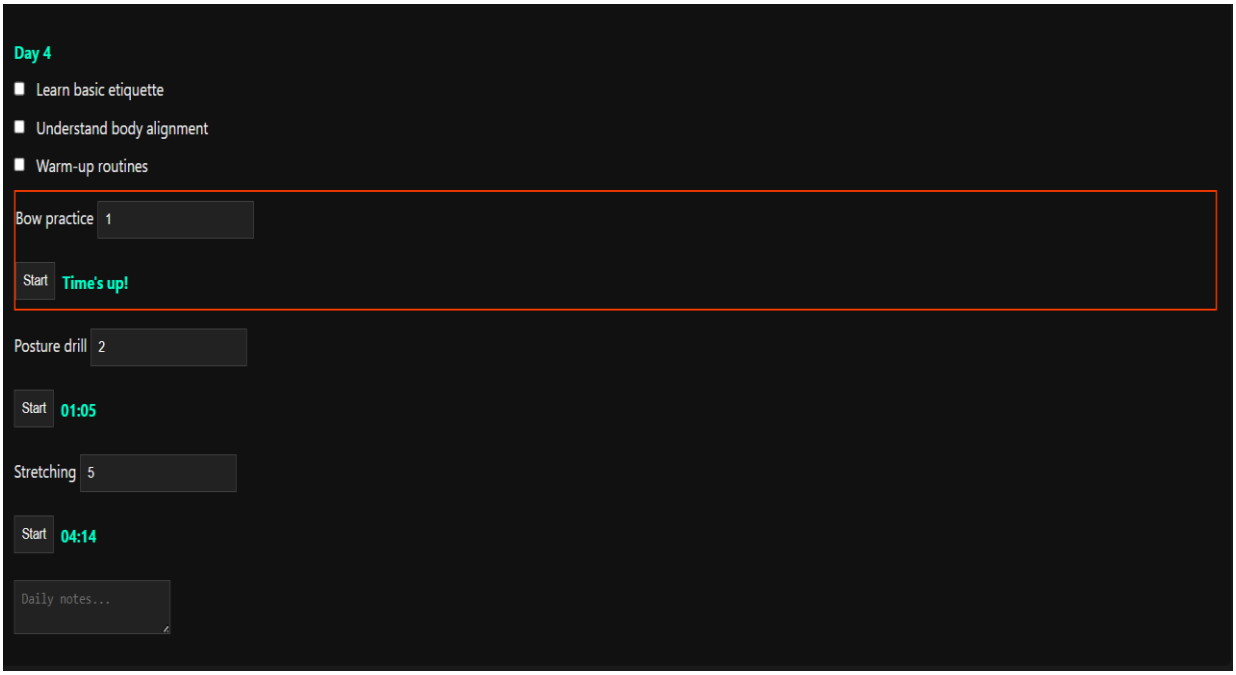
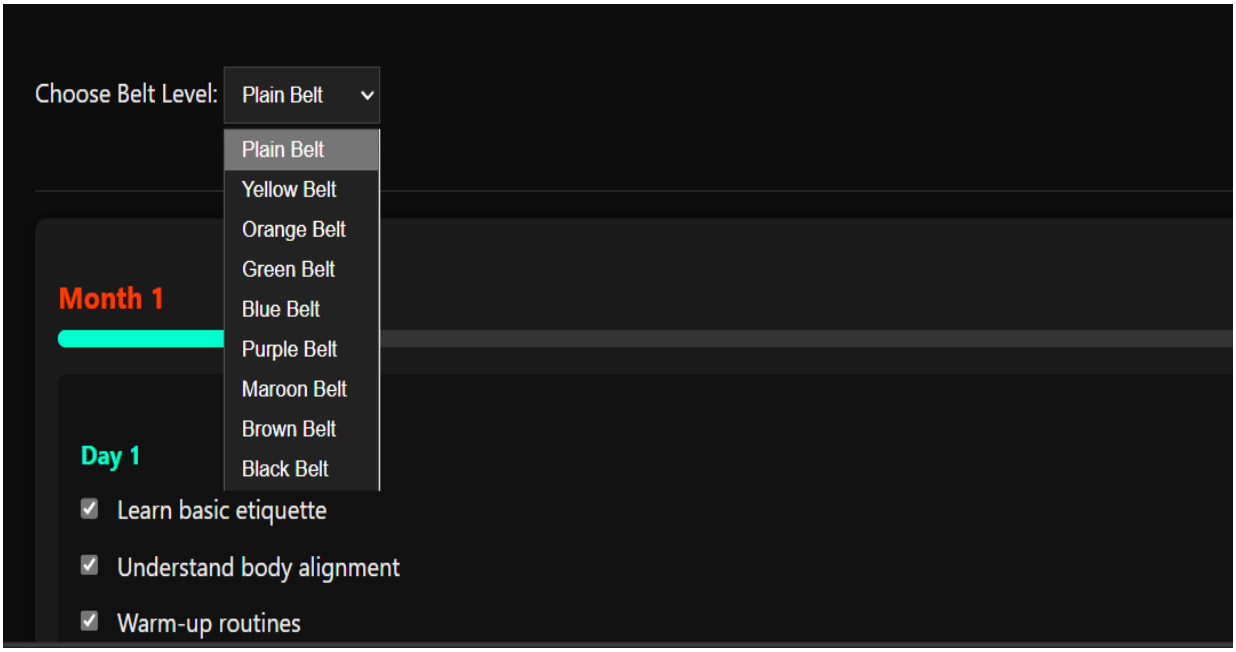
4. Martial Track Menu



5. Progress Tracker Bar



6. Timer



7. Choosing Level